Q. No.	Questions
UNIT 1	**SECTION 1**
1	A mobile manufacturing company is comparing Android with a proprietary mobile operating system before selecting a platform for developing next-generation enterprise applications that require hardware integration, application interoperability, and efficient memory management. Analyze the suitability of Android as a development platform by relating its SDK features, Dalvik Virtual Machine, and open application framework to the given requirements. (5 Marks)
2	A software company is organizing a three-day Java training program for newly recruited engineers. On the first day, several trainees report different installation issues. Some have installed an Eclipse package meant for another programming language, while others have ignored the license agreement windows, causing the installation to stop unexpectedly. Analyze the installation problems faced by the trainees and recommend suitable solutions to ensure successful Eclipse installation. Justify your recommendations. (5 Marks)
3	A multinational software company plans to standardize a single Java development environment for employees working on Windows, Linux, and macOS. The management is considering Eclipse as the common development platform. Analyze whether Eclipse is an appropriate choice for the organization by discussing five features that support your recommendation. (5 Marks)
4	(i) A public transportation department plans to develop an Android application named Smart Transit to help commuters track buses in real time. The application should continuously obtain the user's location, display nearby buses on a map, provide route updates even when running in the background, and securely store recently searched routes for offline access. Analyze and recommend any three Android SDK features required to implement the Smart Transit application. Justify the role of each selected feature. (3 Marks)
	(ii) A public transportation department plans to develop an Android application named Smart Transit to help commuters track buses in real time. The application should continuously obtain the user's location, display nearby buses on a map, provide route updates even when running in the background, and securely store recently searched routes for offline access. Infer the Background Services that are more suitable than executing the application only in the foreground for providing continuous route updates. (2 Marks)
UNIT 1	**SECTION 2**
5	An educational institution is establishing a new Android application development laboratory with 60 computers. The administrator wants every system to have an identical Android Studio environment so that students experience a consistent development platform. Design a standard installation workflow for deploying Android Studio on all laboratory systems and justify the importance of each major installation step. (5 Marks)

6	An Android developer successfully installs Android Studio but is unable to create and run a virtual Android device during the initial setup. The project deadline is approaching, and the issue must be resolved quickly. Analyze the possible reasons for the emulator setup failure and recommend suitable corrective actions to complete the Android Studio configuration. (5 Marks)
7	A courier company develops an Android parcel tracking application that is tested on different smartphones. The development team notices that the user interface appears correctly on some devices but overlaps on others because of varying screen sizes and hardware specifications. They decide to validate the application using Android Virtual Devices before deployment. Analyze Android Virtual Devices (AVDs) in assisting developers to test applications across different hardware configurations and justify the importance of configuring appropriate AVD parameters. (5 Marks)
8	(i) A hospital develops an Android application that sends appointment reminders. Developers notice that reminder messages are not delivered after the device is restarted unless the application is manually opened again. Identify the Android component used to respond automatically after device startup and state its role. (3 Marks)
	(ii) A tourism application has been developed for visitors using high-resolution smartphones. During deployment, users with older devices report that the application is unavailable in the Play Store because of hardware compatibility restrictions. Suggest two modifications that would increase device compatibility without changing the application's core functionality. (2 Marks)
UNIT 2	**SECTION 1**
9	A multinational retail company plans to develop an Android application that enables customers to browse products, purchase items, track orders, receive promotional notifications, and manage their profiles.The development team has been instructed to divide the application into independent modules so that new features can be incorporated without affecting the existing functionalities.The software architect must recommend an appropriate Android architecture that supports modularity, scalability, and efficient maintenance. Analyze the suitability of Android's component-based architecture for developing the above application.Determine the role of each Android application component in satisfying the functional requirements of the system and justify your recommendations. (5 Marks)
10	An online shopping application allows customers to browse products, add items to the cart, make payments, and view order history.During testing, the development team observes that the application becomes slow after users repeatedly navigate between multiple screens.Memory usage also increases continuously, affecting application performance. Analyze the possible factors contributing to the observed performance issue. Recommend suitable Activity lifecycle methods that should be managed to improve memory utilization and ensure efficient application execution. (5 Marks)
11	A music streaming application continues playing songs while users browse other applications, lock the device screen, or switch between different mobile applications.The development team reviews the application's architecture to identify the most appropriate Android application component for implementing

	this functionality. The following Android application components are available: Activity Service Broadcast Receiver Content Provider AndroidManifest.xml The following information is provided: Performs long-running operations. Executes in the background. Does not provide a graphical user interface. Supports continuous music playback. Improves multitasking. Select the most suitable Android application component for the above application and justify your selection using at least four points. (5 Marks)
12	(i) A food delivery application sends an Intent to open the Order Details screen.The Intent contains the Order ID, Customer Address, and Payment Status so that the receiving Activity can display the complete order information. (a) Select the two Intent components required to transfer the above information from the following options. i.Intent Extras ii.Intent Filter iii.Intent Action iv.Intent Category (3 Marks)
	(ii) A food delivery application sends an Intent to open the Order Details screen.The Intent contains the Order ID, Customer Address, and Payment Status so that the receiving Activity can display the complete order information. (b) Specify the purpose of Intent Extras in the above application. (2 Marks)
UNIT 2	**SECTION 2**
13	A hospital develops an Android application to manage patient registration, doctor appointments, laboratory reports, medicine reminders, and emergency notifications. During testing, the project manager observes that different functionalities require different execution mechanisms. Some tasks require direct user interaction, whereas others must continue executing even when the application is running in the background. Selecting inappropriate Android components may reduce application performance and affect user experience. Identify the functional requirements of the hospital application and recommend suitable Android components for each requirement. Justify your recommendations based on the execution characteristics of Android application components. (5 Marks)
14	A national online banking organization is developing an Android application that enables customers to transfer funds, pay utility bills, view transaction history, and receive fraud alerts. During the design review, the software architect recommends separating the user interface from background operations to improve responsiveness. The development team must determine the most suitable Android components for each

	functionality while ensuring secure and efficient application execution. Determine the Android components required for implementing the banking application. Assess how the selected components improve application performance, user experience, and maintainability. (5 Marks)
15	A logistics company develops an Android application that continuously tracks delivery vehicles during transportation.Developers notice that the tracking service continues running even after the delivery is completed, resulting in unnecessary battery consumption.The software team reviews the lifecycle of the Service before deploying the application. Construct the lifecycle of the Service for the above application and analyze how proper lifecycle management improves application performance. (5 Marks)
16	(i) A news organization develops an Android application that should launch automatically whenever users select a news article link from a web browser.During testing, the application installs successfully but never appears in the list of applications available to open the selected link.The development team reviews the Intent Filter configuration. Identify three possible reasons why the Android application is not selected to handle the incoming Intent. (3 Marks)
	(ii) A news organization develops an Android application that should launch automatically whenever users select a news article link from a web browser.During testing, the application installs successfully but never appears in the list of applications available to open the selected link.The development team reviews the Intent Filter configuration. (b) Indicate any two Intent Filter components required for successful Intent matching. (2 Marks)
UNIT 3	**SECTION 1**
17	A retail inventory application stores thousands of product records, including product ID, name, price, and available quantity. The development team plans to organize all database operations using a structured approach that simplifies database creation, upgrades, and maintenance. The software architect reviews the database design before implementation. Construct the SQLite database architecture suitable for the above application and indicate the role of SQLiteOpenHelper in database management. (5 Marks)
18	A supermarket Android application records product details in an SQLite database, but newly added products are not reflected in the inventory list despite successful save confirmation. The development team examines the database interaction to identify the source of the inconsistency. Analyze the database interaction and distinguish the SQLite operations required for inserting, querying, and displaying product records, relating each operation to the observed inventory issue. (5 Marks)
19	A university develops an Android application to maintain student internship records.During software testing, faculty members report that student records can be added and viewed successfully, but updating internship company details and removing completed internship records do not work as expected.The development team reviews the database operations implemented in the application. Analyze the database operations performed in the application and identify the SQLite CRUD operation associated with each of the following activities.Indicate how the identified operations support effective database management. 1.Registering a new internship record

	2.Displaying internship details 3.Modifying internship company information 4.Removing completed internship records (5 Marks)
20	(i) An online examination application allows students to answer multiple-choice questions and choose their preferred examination language.During testing, developers observe that the selected answer disappears whenever the screen rotates, while the selected examination language should remain available even after the application is closed and reopened.The development team reviews the Android data storage APIs before deployment. (a) Investigate the above scenario and identify the Android data storage API that should be used to preserve the selected answer during screen rotation.Indicate why the selected API is appropriate. (3 Marks)
	(ii) An online examination application allows students to answer multiple-choice questions and choose their preferred examination language.During testing, developers observe that the selected answer disappears whenever the screen rotates, while the selected examination language should remain available even after the application is closed and reopened.The development team reviews the Android data storage APIs before deployment. (b) Classify the following application data based on the appropriate Android data storage API. 1) Selected examination answer 2) Preferred examination language (2 Marks)